

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: CSA5113

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO4: *Implement best security practices for IP, email, and web service using PGP, S/MIME for enhancing email Security.CO (BL3, BL6)*

Assessment Tool Weightage: 10%

QUESTIONS: 5 TOTAL MARKS: 50

Annexure A

Comparative Analysis

Code of Conduct:

I DINDU PUJITHA(192373037) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: D PUJITHA

Annexure A
Comparative Analysis

1. Compare Authentication Header (AH) and Encapsulating Security Payload (ESP) in IPSec with respect to security services, performance overhead, and suitability for different network scenarios. Analyze which is more appropriate for a VPN deployment.

Answer:

Compare Authentication Header (AH) and Encapsulating Security Payload (ESP) in IPSec

Feature	AH	ESP
Full form	Authentication Header	Encapsulating Security Payload
Confidentiality	No	Yes
Data integrity	Yes	Yes
Authentication	Yes	Yes
Anti-replay protection	Yes	Yes
Encryption	No	Yes
Performance overhead	Lower because there is no encryption	Higher because encryption and authentication are performed
NAT compatibility	Poor	Better

Common usage	Integrity/authentication when encryption is not required	VPNs and secure communication
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Analysis:

AH protects the integrity and authenticity of IP packets but does not encrypt the actual data. Therefore, anyone who can observe the network may still see the contents of the communication.

ESP provides encryption along with integrity, authentication, and anti-replay protection. It therefore provides stronger overall security.

Best choice for VPN:

ESP is more appropriate for VPN deployment. VPNs normally require confidentiality as well as integrity and authentication. ESP can encrypt the transmitted data, preventing attackers from reading sensitive information.

Conclusion:

ESP is generally preferred over AH for modern VPNs because it provides confidentiality and works better with common NAT-based network environments.

2. Compare transport mode and tunnel mode in IPSec architecture. Evaluate their effectiveness in securing host-to-host versus network-to-network communication.

Answer:

Feature	Transport Mode	Tunnel Mode
I P header	Original IP header remains	Original IP packet is encapsulated
Protection	Mainly protects the payload	Protects the entire original IP packet
New IP header	No	Yes
Overhead	Lower	Higher
Performance	Faster	Slightly slower
Host-to-host	Highly suitable	Can also be used
Network-to-network	Less suitable	Highly suitable
VPN usage	Limited	Very common

Transport Mode:

In transport mode, the original IP header remains visible while the payload is protected.

Original IP Header + Protected Data

It is mainly useful for **host-to-host communication**, where individual computers communicate directly.

Tunnel Mode:

In tunnel mode, the complete original IP packet is protected and placed inside a new IP packet.

New IP Header + Protected Original IP Packet

It is commonly used for **network-to-network VPNs**, such as connecting two company offices over the Internet.

Comparison:

For **host-to-host communication**, transport mode is efficient because it has lower overhead.

For **network-to-network communication**, tunnel mode is more effective because it protects the complete original packet and hides the internal network addresses.

Conclusion:

- **Host-to-host → Transport Mode**
- **Network-to-network VPN → Tunnel Mode**

Tunnel mode is generally preferred for corporate VPNs.

- 3. Perform a comparative analysis of Pretty Good Privacy (PGP) and S/MIME in terms of key management, trust models, scalability, and enterprise usability. Recommend the most suitable approach for a corporate email system.**

Answer:

Feature	PGP	S/MIME
Full form	Pretty Good Privacy	Secure/Multipurpose Internet Mail Extensions
Key management	User-controlled	Usually certificate-based
Trust model	Web of Trust	Certificate Authority (CA)
Scalability	More difficult for large organizations	Better for enterprises
Email clients	Supported by many clients through extensions	Widely supported by enterprise email clients
Digital signatures	Yes	Yes
Encryption	Yes	Yes
Enterprise management	More complex	Easier
Centralized administration	Limited	Strong

PGP:

PGP uses a **Web of Trust** model. Users can generate and manage their own keys and can verify or sign other users' keys.

Advantages:

- Flexible key management
- Does not depend entirely on a central CA
- Strong encryption and digital signatures

Disadvantages:

- Key management can become complicated
- Difficult to manage for large organizations
- Requires users to understand key verification and management

S/MIME:

S/MIME uses **digital certificates** issued by trusted Certificate Authorities.

Advantages:

- Centralized certificate management
- Easier deployment in organizations
- Good integration with enterprise email systems
- Easier to manage large numbers of employees

Disadvantages:

- Depends on certificate infrastructure
- Certificate issuance and renewal require management

- Trust is dependent on the CA infrastructure

Recommendation:

For a **corporate email system**, **S/MIME is generally the better choice** because it provides centralized certificate management, scalability, and strong integration with enterprise email environments.

Conclusion:

PGP is suitable when decentralized trust and user-controlled keys are preferred, while **S/MIME is more suitable for large corporate environments**.

4. Compare rule-based and machine learning-based email spam detection techniques. Analyze their effectiveness in handling evolving spam patterns and minimizing false positives.

Answer:

Feature	Rule-Based	Machine Learning-Based
Working method	Uses predefined rules	Learns patterns from data
Training required	No	Yes
Detects known spam	Good	Good
Detects new spam patterns	Limited	Better
Adaptability	Low	High
False positives	Can be high if rules are too strict	Can be reduced with good training
Maintenance	Rules must be updated manually	Model can be retrained
Complexity	Low	Higher
Scalability	Limited for evolving threats	Better

Rule-Based Detection

Rule-based systems identify spam using predefined conditions.

Examples:

IF message contains "WIN MONEY"

THEN classify as SPAM

or:

IF sender is on blacklist

THEN classify as SPAM

Advantages

- Simple to implement
- Easy to understand
- Fast
- Does not require training data

Disadvantages

- Cannot easily detect new spam techniques
- Spammers can modify messages to bypass rules
- Too many strict rules may cause legitimate emails to be classified as spam

Machine Learning-Based Detection

Machine learning systems learn characteristics of spam from previously classified emails.

Common algorithms include:

- Naive Bayes
- Decision Trees
- Random Forest
- Support Vector Machines
- Neural Networks

Advantages

- Can detect changing spam patterns
- Adaptable to new types of spam
- Can reduce false positives when trained properly

- Suitable for large-scale email systems

Disadvantages

- Requires training data
- Model needs monitoring and retraining
- More complex than rule-based systems

Conclusion

Machine learning-based spam detection is generally more effective for modern email systems because spam patterns continuously change. A hybrid approach combining **rules + machine learning** can provide even better protection.

5. **Compare manual key management and automated key management (e.g., using IKE) in IPSec. Evaluate their impact on security, scalability, and operational complexity in large organizations.**

Answer:

Feature	Manual Key Management	Automated Key Management (IKE)
Key distribution	Manually configured	Automatically negotiated
Scalability	Poor	High
Operational complexity Operational complexity	Very high	Lower
Key rotation	Manual	Automatic
Large organizations	Not suitable	Highly suitable

Configuration effort	High	Lower after initial setup
Human errors	More likely	Reduced
Dynamic environments	Poor	Excellent

Manual Key Management

In manual key management, administrators configure encryption keys and security parameters on each IPSec device.

Example:

Router A → Manually configured key

Router B → Same manually configured key

For a small network, this can work reasonably well.

However, as the number of devices increases, managing keys becomes difficult.

Problems

- Keys must be manually configured
- Key changes require administrative work
- Greater possibility of configuration errors
- Difficult to manage hundreds or thousands of devices
- Manual key rotation is complicated

Automated Key Management Using IKE

IKE (Internet Key Exchange) automatically negotiates security parameters and establishes shared keys between IPSec peers.

It provides:

- Automated key exchange
- Authentication
- Security association negotiation

- Key renewal/rekeying
- Better scalability

Large Organization Example

Suppose an organization has **500 VPN connections**.

With manual key management:

500 connections



Manual configuration



High administrative effort



Higher chance of errors

With IKE:

500 connections



Automatic negotiation



Automatic key establishment/rekeying



Better scalability

Conclusion

Automated key management using IKE is much more suitable for large organizations.

It improves scalability, reduces administrative workload, supports automatic key establishment and rekeying, and reduces the possibility of human configuration errors.